

Naser Souri

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🎓 Scholar

🌐 LinkedIn

Education

Virginia Polytechnic Institute and State University

Ph.D. in Electrical Engineering

August 2022 – Present

GPA: 3.8/4.0

Dissertation: Control and Stability Enhancement for the Integration of Power Converters Into the Grid

University of Tehran

M.Sc. in Electrical Engineering

August 2016 – September 2019

GPA: 3.8/4.0

Thesis: Maximum Energy Harvesting in Solar Panels Under Partial Shading Conditions in DC Grids

Arak University of Technology

B.Sc. in Electrical Engineering

August 2011 – September 2015

GPA: 3.7/4.0

Final Project: Design and Implementation of a Single-Phase Inverter and DC-DC Converter

Work Experience

Virginia Tech (Instructor)

August 2026 – December 2026

Instructor

Blacksburg, Virginia

- AC circuit analysis for electrical engineering major

Claros (Internship)

July 2026 – August 2026

Power Electronics Control Engineer

California

- Develop system-level models for large data centers, including 800-V DC architectures
- Perform EMT, power-quality, and grid-interaction studies using PSCAD/MATLAB
- Conduct large-load transient and stability studies

GE Vernova (Internship)

May 2025 – December 2025

Power Electronics Control Engineer

Albany, New York

- Developed grid-forming (GFM) and grid-following (GFL) inverter control strategies
- Evaluated and simplified GFM/GFL control algorithms for efficient firmware implementation
- Implemented Python scripts for AEMO grid code compliance testing
- Automated post-processing of inverter waveforms for grid code compliance using Python

Virginia Polytechnic Institute and State University

August 2022 – Present

Graduate Research Assistant

Blacksburg, Virginia

- Developed and validated a real-time model for a Google data center using RSCAD/RTDS
- Accelerated electromagnetic transient (EMT) simulations for large-scale data centers in PSCAD
- Analyzed stability and control of power converters in AC and DC grids
- Implemented resonance damping techniques on a three-phase grid-connected inverter using PHIL
- Designed control strategies for improved dynamic power sharing in microgrids
- Developed circulating current suppression strategies and improved microgrid efficiency

Vehicle, Fuel, and Environment Research Institute

February 2020 – March 2021

Power Electronics Engineer

Tehran, Iran

- Developed a controller for auxiliary inverters for electric buses

- Contributed to the auxiliary inverter circuits, hardware, and PCB design

University of Tehran

Graduate Research Assistant

August 2017 – September 2019

Tehran, Iran

- Contributed to the design of an 11 kW electric motor drive converter
- Implemented a controller for a 2 kW DC–DC converter integrated with an inverter
- Designed, simulated, and implemented a 3 kW induction heating system
- Designed and simulated a 1 kW dual active bridge (DAB) DC–DC converter

Arak University of Technology

Research Assistant

February 2015 – September 2015

Arak, Iran

- Designed and simulated a 600 W standalone inverter integrated with a boost DC–DC converter
- Developed inverter and boost converter hardware and PCB layouts
- Implemented the P&O MPPT algorithm for the boost converter
- Programmed standalone inverter control algorithms on a microcontroller

Niro Research Institute (NRI)

Power Electronics Engineer

October 2015 – July 2016

Tehran, Iran

- Contributed to the design and implementation of a 17 MVAR static VAR compensator (SVC)
- Designed and tested thyristor gate-drive circuits for the SVC
- Performed high-voltage equipment assembly and cable routing for the SVC
- Developed variable-speed drive control strategies for induction and permanent-magnet motors

Technical Skills

Power Electronics	Interleaved buck converters, LLC/CLLC resonant converters, SiC/GaN devices, EMI/EMC
Data Center & BESS	BBU/CBU, busbars, 48-V/800-V DC architectures, BESS/PCS control, OEM/black-box PSCAD models
Grid Interconnection	Working knowledge of NERC/PJM requirements
Programming Languages	Python (NumPy, Pandas, SciPy), C, C++, L ^A T _E X
Simulation Software	MATLAB/Simulink, PSCAD, RSCAD/RTDS, EMTP, SIMPLIS, PSS [®] E, PLECS, PSIM, PSpice, COMSOL, Altium
Hardware-in-the-Loop (HIL)	RTDS, Typhoon HIL 402/404
Laboratory Tools	Oscilloscope, Power Analyzer, Megger, Autotransformers, Function Generator, etc.
Productivity Tools	Microsoft 365, Overleaf, Jama, Jira, Confluence
Embedded Systems	ARM, PIC, dsPIC, DSP

Peer Reviewer (100+ Reviews)

IEEE Transactions on Industrial Electronics
 IEEE Transactions on Energy Conversion
 IET Generation, Transmission & Distribution
 Journal of Modern Power Systems and Clean Energy
 Iranian Journal of Science and Technology, Transactions of Electrical Engineering
 IEEE Applied Power Electronics Conference (APEC)
 IEEE Power & Energy Society General Meeting (PESGM)
 IEEE Industrial Electronics Society Annual Conference (IECON)
 IEEE Innovative Smart Grid Technologies (ISGT)
 North American Power Symposium (NAPS)

Virginia Tech
University of Tehran
Arak University of Technology

Microgrid, Applied Electrical Theory, Electrical Theory
Power Electronics I, Power Electronics II
Electronic Circuits I and II, Electrical Machines, and Engineering Mathematics

Publications

- **N. Souri** “Transient Grid Power Smoothing of Rapid IT-Load Variations in AI Data Centers,” *Submitted for peer review at IEEE Applied Power Electronics Conference and Exposition (APEC)*, New Orleans, LA, March 2027.
- A. Rafee, Y. Batmani, **N. Souri** and A. Mehrizi-Sani, “Harmonic Mitigation and Switching Frequency Reduction in LCL-Filtered Grid-Tied Converters,” *IEEE Transactions on Power Electronics*, 2026. DOI: 10.1109/TPEL.2026.3697501
- **N. Souri** and A. Mehrizi-Sani, “Hybrid droop-based control for parallel operation of DC-DC converters in a DC microgrid,” *IEEE Transactions on Energy Conversion*, pp. 1–12, October 2025. DOI: 10.1109/TEC.2025.3626451
- M. K. Klidbari, **N. Souri**, Z. S. Habibollahi, and H. Montazeri Hedeschi, “A hybrid three-port topology for urban charging stations,” *IEEE Applied Power Electronics Conference and Exposition (APEC)*, Atlanta, GA, March 2025. DOI: 10.1109/APEC48143.2025.10977273
- **N. Souri** and A. Mehrizi-Sani, “Hybrid machine learning approach for cyberattack mitigation of parallel converters in a DC microgrid,” *IEEE Industrial Electronics Society (IECON)*, Chicago, IL, November 2024. DOI: 10.1109/IECON55916.2024.10905595
- **N. Souri**, A. Mehrizi-Sani, and K. Tehrani, “Stability enhancement of LCL-type grid-following inverters using capacitor voltage active damping,” *IEEE Power & Energy Society General Meeting (PESGM)*, Seattle, WA, July 2024. DOI: 10.1109/PESGM51994.2024.10688899
- **N. Souri** and A. Mehrizi-Sani, “Accurate current sharing in a DC microgrid using modified droop control algorithm,” *IEEE Industrial Electronics Society (IECON)*, Chicago, IL, November 2024. DOI: 10.1109/IECON55916.2024.10905220
- **N. Souri**, S. Farhangi, H. Iman-Eini, and H. Ziar, “Modeling and estimation of the maximum power of solar arrays under partial shading conditions,” *11th Power Electronics, Drive Systems, and Technologies Conference (PEDSTC)*, Tehran, Iran, February 2020. DOI: 10.1109/PEDSTC49159.2020.9088407
- A. Zaheri, M. Farahani, A. Sadeghi, and **N. Souri**, “Transient liquid phase bonding of Cu and Al using metallic particles interlayers,” *Journal of Engineering Manufacture*, vol. 235(9), pp. 1518–1528, 2021. DOI: 10.1109/PESGM51994.2024.10688899